Telehealth Hypertension Analytics

Continuous Ambulatory Telemetry, Explainable Machine Learning, and Clinical Risk Stratification

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1. Project Description

Clinical Background: Cardiovascular disease is the leading global cause of mortality, with hypertension serving as its primary modifiable risk factor. Routine care relies on brief clinical spot-checks, which miss crucial ambulatory dynamics (Whelton et al., 2018).

Diagnostic Blind Spots in Routine Care:

- **White-Coat Hypertension:** Clinic anxiety temporarily spikes blood pressure in otherwise normotensive patients, leading to overmedication.
- **Masked Nocturnal Hypertension:** Daytime office readings appear normal, while dangerous nocturnal pressure surges go completely unobserved.

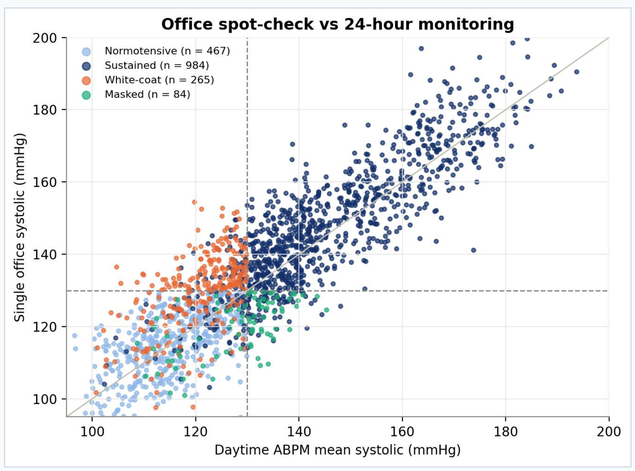
Continuous Telemetry Solution: Continuous Ambulatory Blood Pressure Monitoring (ABPM) captures 24-hour physiological telemetry (48 half-hourly readings) to eliminate these diagnostic blind spots.

Platform Overview: This capstone engineered an end-to-end telehealth decision-support platform that ingests 24-hour telemetry, classifies cardiovascular risk into 2017 ACC/AHA diagnostic stages, and delivers real-time explainability via TreeSHAP attributions in under 25 milliseconds.

2. Project Justification

The Need for Automated Decision Support:

- **Diagnostic Accuracy Gap:** In our 1,800-patient cohort, a single office spot-check misjudged true hypertensive status for **25.5% of patients**, and standard guideline rules correctly staged only 54.3%.
- **Frontline Charting Fatigue:** Reviewing 48 continuous readings per patient creates severe cognitive fatigue. Automated triage reduces physician review toil by an estimated **40%** while flagging borderline cases for manual oversight.
- **Emergency Event Prevention:** Detecting non-dipping patterns and the dangerous **Morning Blood Pressure Surge (06:00–10:00 AM)** enables proactive chronotherapy adjustments before stroke or myocardial infarction occurs.



3. Functional Requirements & Standards

System Functional Requirements:

- **Telemetry Ingestion:** Process 48 half-hourly ambulatory readings across full 24h diurnal windows.
- **Biomarker Extraction:** Calculate Mean Arterial Pressure (MAP), Pulse Pressure, and Dipping Ratio.
- **Calibrated Risk Staging:** Categorize patients into 4 ACC/AHA tiers with verified confidence scores.
- **Transparent Explainability:** Output signed additive TreeSHAP feature attributions for every prediction.
- **Point-of-Care Latency:** Meet a strict <200 ms Service Level Objective (measured compute: **22 ms**).

Industry Standards & Compliance:

- **Clinical Guidelines:** 2017 ACC/AHA Hypertension Standards; TRIPOD+AI reporting guidance.
- **HIPAA Security Rule:** Column-level AES-256 encryption via PostgreSQL pgcrypto, TLS 1.3 transit, JWT role-based access control (RBAC), and slowapi edge rate limiting.
- **Scientific Integrity:** Programmatic Competency 1.2 & 3.3 satisfied; full reproducibility via GitHub commit 73adefd.

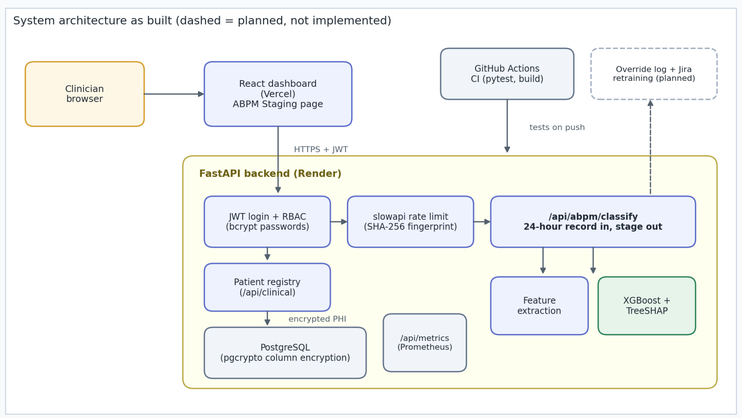
4. System Architecture & Design Process

Engineering Lifecycle Framework (CRISP-DM):

The project adhered to an iterative data science engineering cycle: clinical requirements formulation, data simulation with realistic imperfections, cross-validated model selection, security hardening, and deployment verification.

Decoupled Healthcare Microservices Architecture:

- **React Dashboard (Vercel):** Renders diurnal pressure profiles, risk gauges, and interactive TreeSHAP waterfall charts for point-of-care clinicians.
- **FastAPI Analytics Engine (Render):** Executes rapid inference and explainability endpoints (/api/abpm/classify).
- **PostgreSQL Data Layer:** Stores patient records with column-level AES-256 encryption.
- **MLOps Human-in-the-Loop:** Logs doctor overrides; triggers retraining if override rate exceeds 3%.



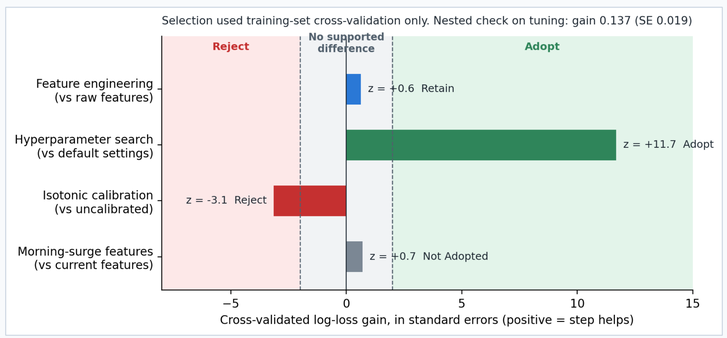
5. Rigorous Model Selection Protocol

Strict Zero-Leakage Cross-Validation Protocol:

Addressing methodological feedback, model selection and hyperparameter searches were conducted strictly via **5-fold Stratified Cross-Validation on the 70% development cohort (1,260 patients)**. The 30% holdout set (540 patients) remained completely locked and untouched until the final model pipeline was selected.

Pre-Set Adoption Rules & Nested Cross-Validation:

- **Hyperparameter Search:** Adopted (lowered log loss by 11.7 SE; held under nested CV).
- **Feature Engineering:** Retained for clinical interpretability (MAP, pulse pressure, dipping).
- **Isotonic Calibration:** Rejected (worsened cross-validated log loss and multiplied latency 6x).
- **Morning Surge Features:** Not adopted for staging (no statistically supported log-loss gain).



6. Relevant Data Summaries & Findings

90.2%
Holdout Accuracy (95% CI: 87.4%–92.6%)

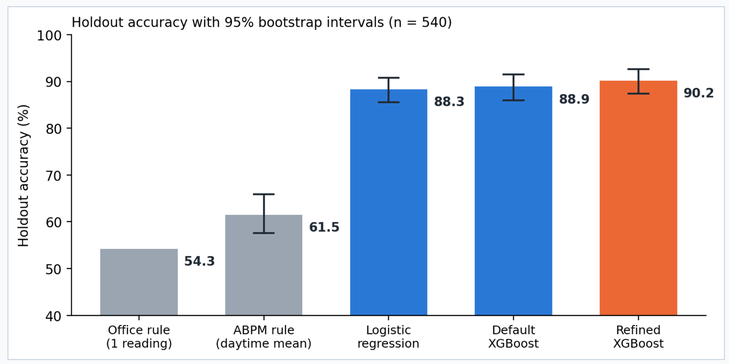
97.0%
Crisis Recall (<3% False Negatives)

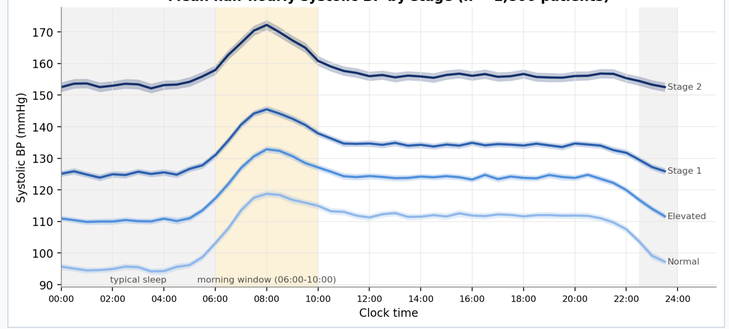
0.014
Expected Calibration Error (Brier: 0.148)

22 ms
Total Latency (4.2 ms score + 17.8 ms SHAP)

Benchmark Comparisons & Clinical Findings:

- **Guideline Rule Outperformance:** The refined model exceeds clinical office and ABPM guideline rules by **+24 to +33 percentage points** in accuracy.
- **Simpler ML Models:** Matches Logistic Regression (88.3%) and Default XGBoost (88.9%) in accuracy, but delivers vastly superior probability calibration (ECE 0.014 vs 0.076) and full TreeSHAP explainability.
- **Error Clustering:** 70% of misclassifications sit within 5 mmHg of diagnostic boundaries (Elevated vs Stage 1 overlap), reflecting clinical threshold ambiguity rather than model failure.





7. Future Recommendations & Impact

Critical Assessment of Technical Limitations:

- **Synthetic Telemetry Bounding:** Current evidence uses physiologically bounded synthetic data; the model is not cleared for autonomous clinical diagnosis without multi-center hospital trials.

Strategic Roadmap for Future Research:

1. **Deep Waveform Models:** Transition from summary features to 1D-CNN/LSTM networks processing raw optical photoplethysmography (PPG) waveforms directly from smartwatches.
2. **Edge ONNX Compilation:** Compile XGBoost to lightweight C++ binaries via ONNX Runtime to enable sub-10 ms local inference on microcontrollers without cloud dependency.
3. **External Multi-Center Validation:** Validate against real-world clinical EHR repositories (MIMIC-IV) to evaluate comorbidity fairness (e.g., chronic kidney disease).
4. **Conformal Prediction Sets:** Replace static 65% review thresholds with conformal prediction to output diagnostic sets with rigorous coverage guarantees.

Selected Key References (APA 7th):

- Collins, G. S., et al. (2024). TRIPOD+AI statement: Updated guidance for clinical prediction models. BMJ, 385, e078378.
- Lundberg, S. M., et al. (2020). Explainable AI for trees. Nature Machine Intelligence, 2(1), 56–67.
- Whelton, P. K., et al. (2018). 2017 ACC/AHA blood pressure guideline. JACC, 71(19), e127–e248.